

Working with Data in R

Reading for Class #8: Thursday, Feb 04, 2027

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Reading:

Required Reading (everyone):

- R for Data Science, Ch. 4–6.

Reading Notes:

Overview of the reading

- See the homework assignment for the exercises you should do.

Chapter 4

This very short chapter goes over guidance on style for naming variables and functions and formatting your code to make it easy to read and keep it organized.

Chapter 5

This chapter introduces principles of tidy data and tools in R for organizing your data into tidy formats.

Principles of Tidy Data

The fundamental idea is that there are many ways to organize a data set in a table with rows and columns. Different kinds of analysis call for different ways of organizing your data. However, for all these ways of organizing, there is an overarching principle, illustrated in the figure below:

1. Each variable is a column and each column is a variable
2. Each observation is a row, and each row is an observation
3. Each value is a cell and each cell is a value

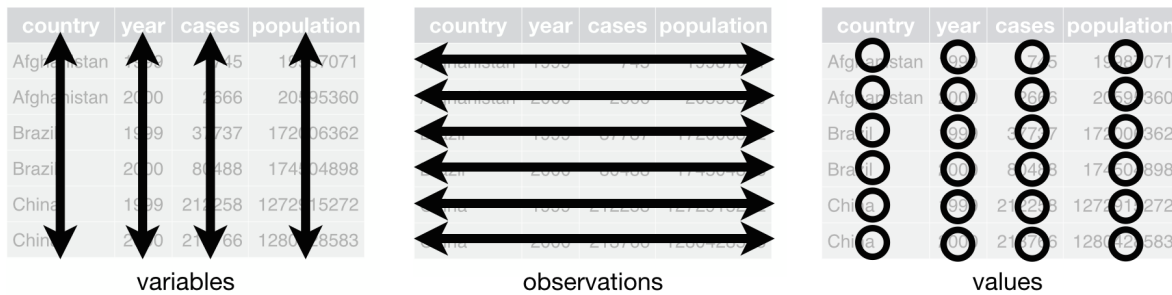


Figure 1: Three rules for making a data set tidy

Pivoting Data

The chapter has two principle parts, each of which focuses on a different aspect of *pivoting* data by rearranging rows and columns.

Lengthening Data The first part looks at tools for *lengthening* data by combining many columns into a single column and adding another column to indicate what each row signifies.

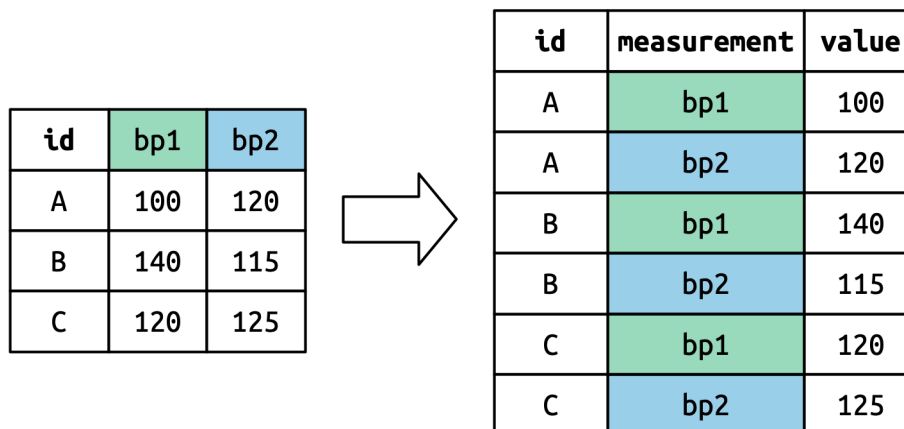


Figure 2: Pivoting to lengthen a data frame by combining two columns into one

The key thing to pay attention to in this reading is the `pivot_longer()` function. The `distinct()` function is also important, because it helps us eliminate redundant rows.

Focus on sections 5.3.1 and 5.3.2.

Section 5.3.3 gets into fancier ways to lengthen data, which you will probably not need, so you can skim this section lightly.

However, do pay enough attention to section 5.3.3 that you will know what is possible and can return to it later if you're reading in complicated data and need the more powerful tools to organize it.

Widening Data The second part looks at tools for *widening* data by splitting up one column into multiple columns. *Widening* is much simpler than *lengthening* data, so this section is shorter. The key thing here is to become familiar with the `pivot_wider()` function.